

# Introduction to Game Theory

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ECON 5001

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# What is Game Theory?

*Game theory* is the study of strategic interaction: settings where each player's best choice depends on what the others do.

- Players, strategies, and payoffs
- Solution concepts: dominance, Nash equilibrium, and refinements
- Applications across economics, political science, biology, and CS

## A First Example: The Prisoner's Dilemma

Two players simultaneously choose to *Cooperate* or *Defect*:

|          |   | Player 2      |                       |
|----------|---|---------------|-----------------------|
|          |   | C             | D                     |
| Player 1 | C | -1, -1        | -3, <u>0</u>          |
|          | D | <u>0</u> , -3 | <u>-2</u> , <u>-2</u> |

*Defect* strictly dominates *Cooperate* for both players, so the unique Nash equilibrium is (D, D) — even though both prefer (C, C).

# Roadmap

1. Strategic-form games and dominance
2. Nash equilibrium
3. Mixed strategies
4. Extensive-form games and backward induction
5. Repeated games and reputation